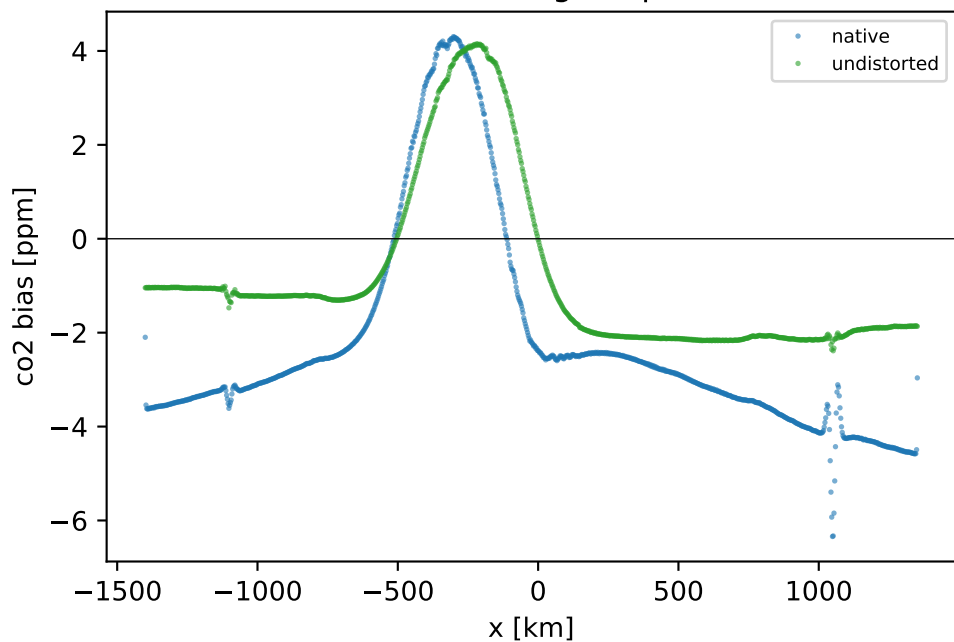
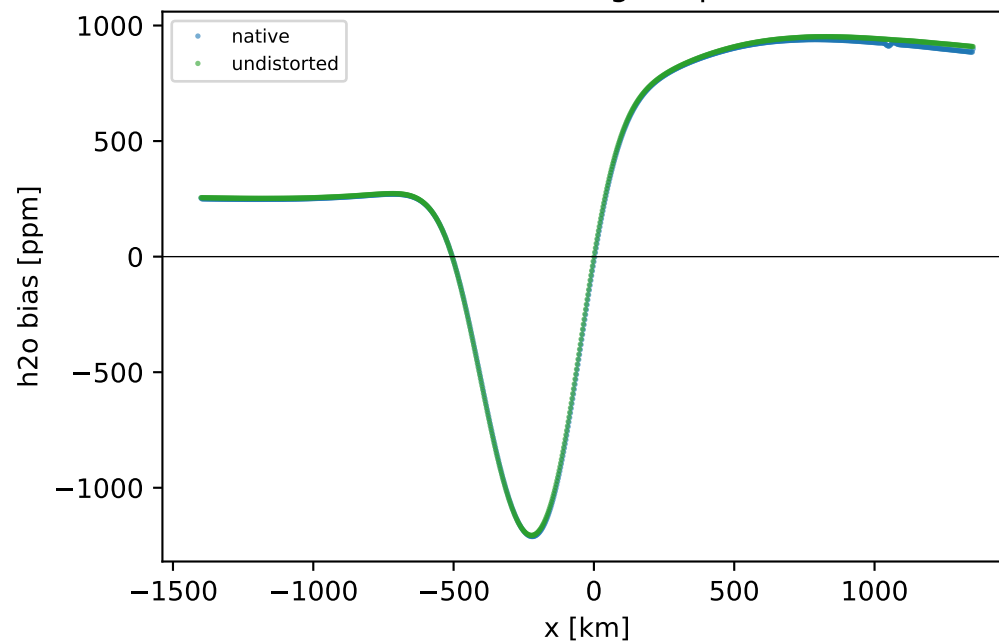


Joint FPA2 -- realistic, no noise (native/undistorted)

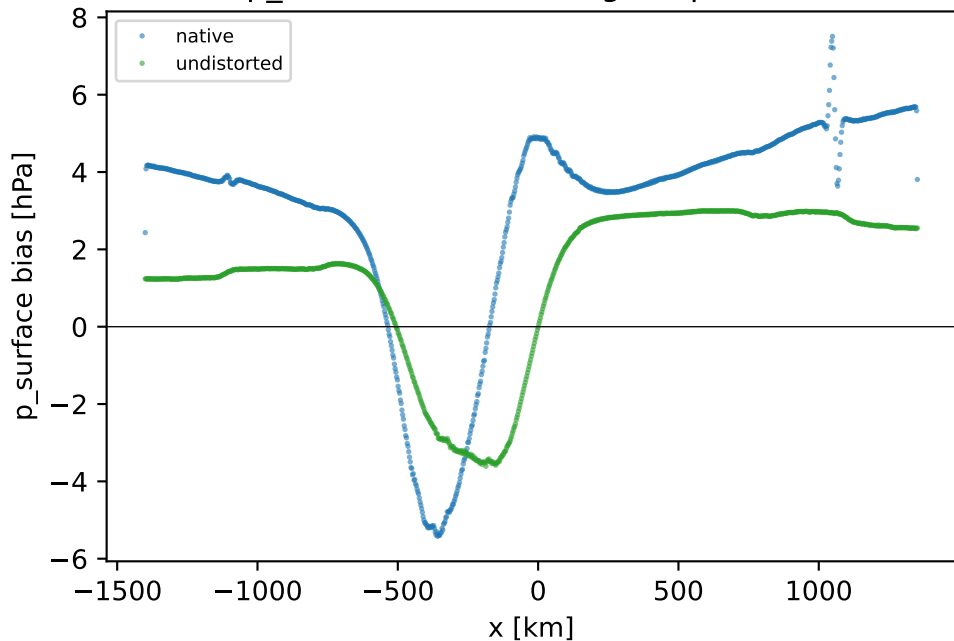
co2 bias vs along-slit position



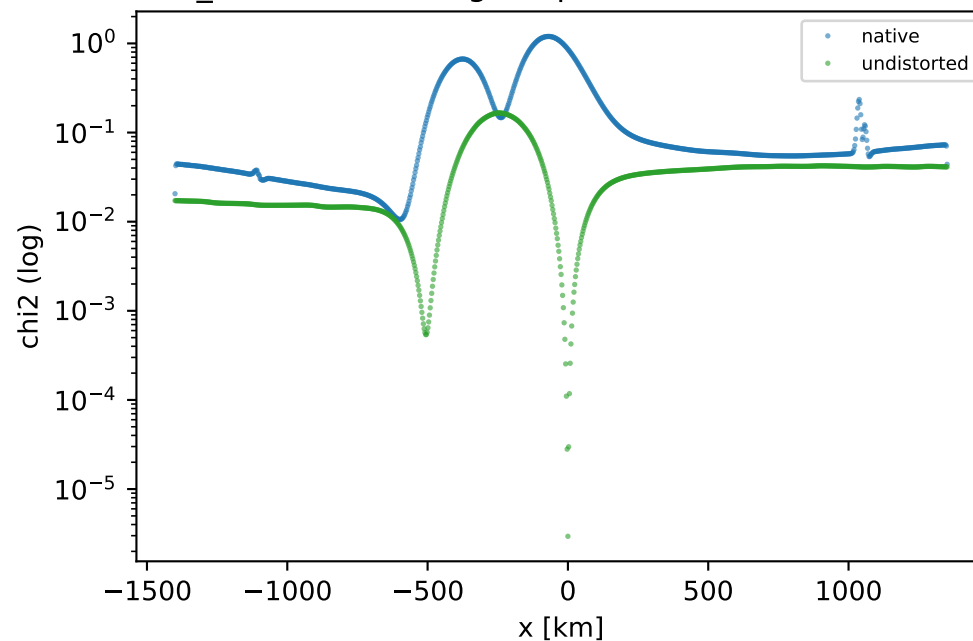
h2o bias vs along-slit position



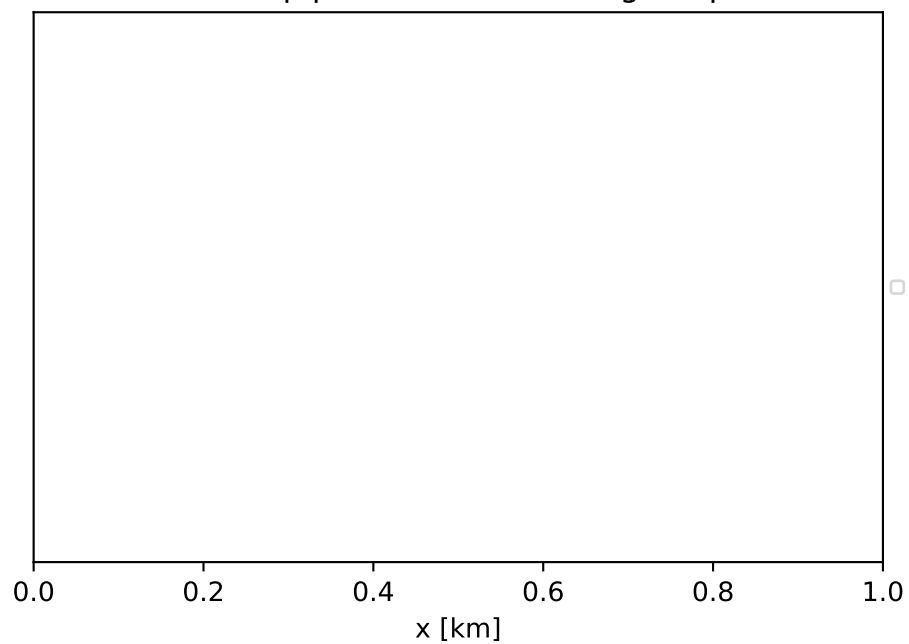
p_surface bias vs along-slit position



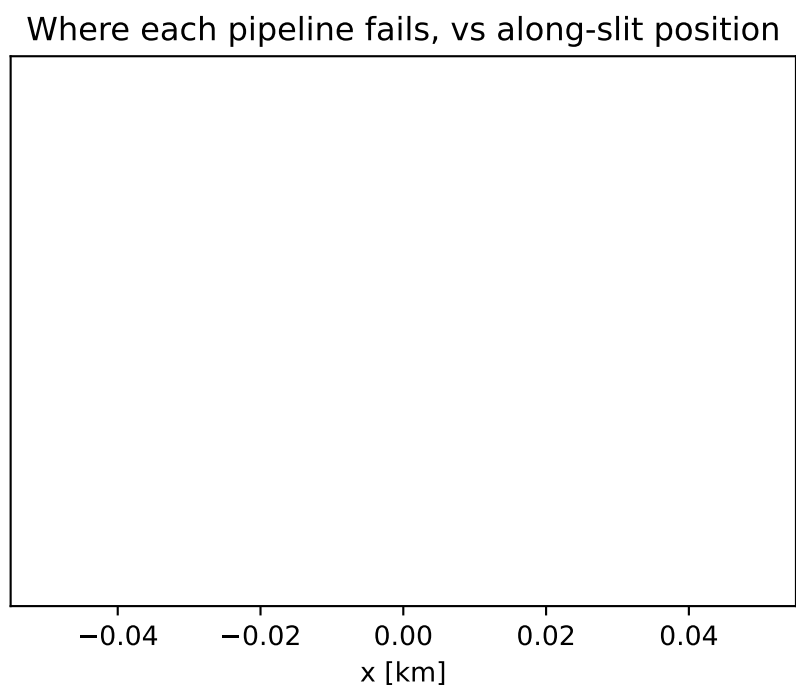
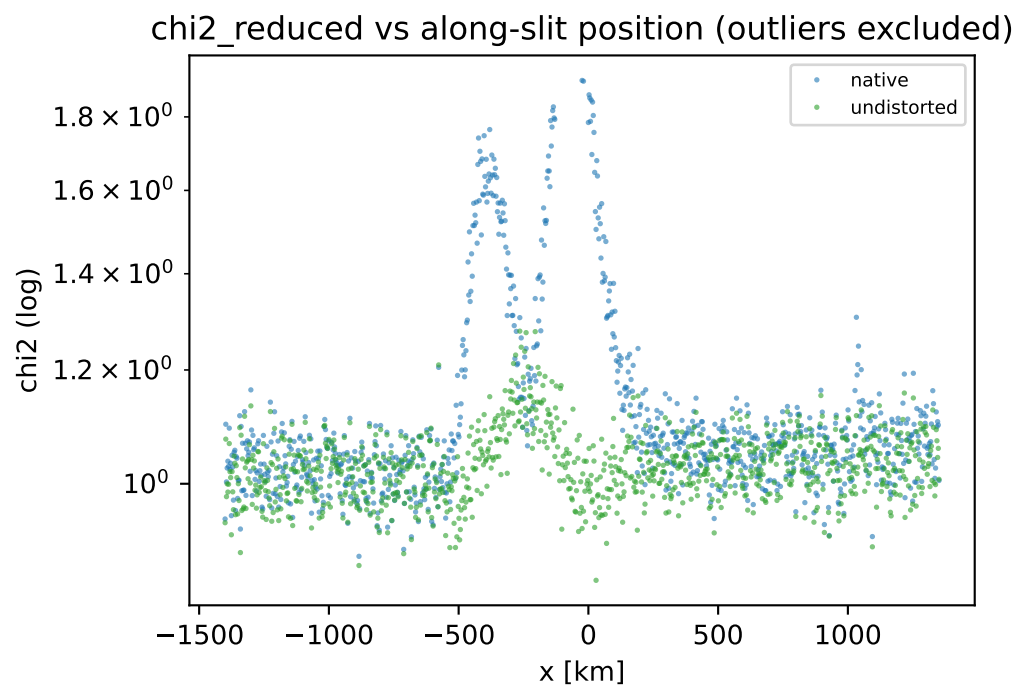
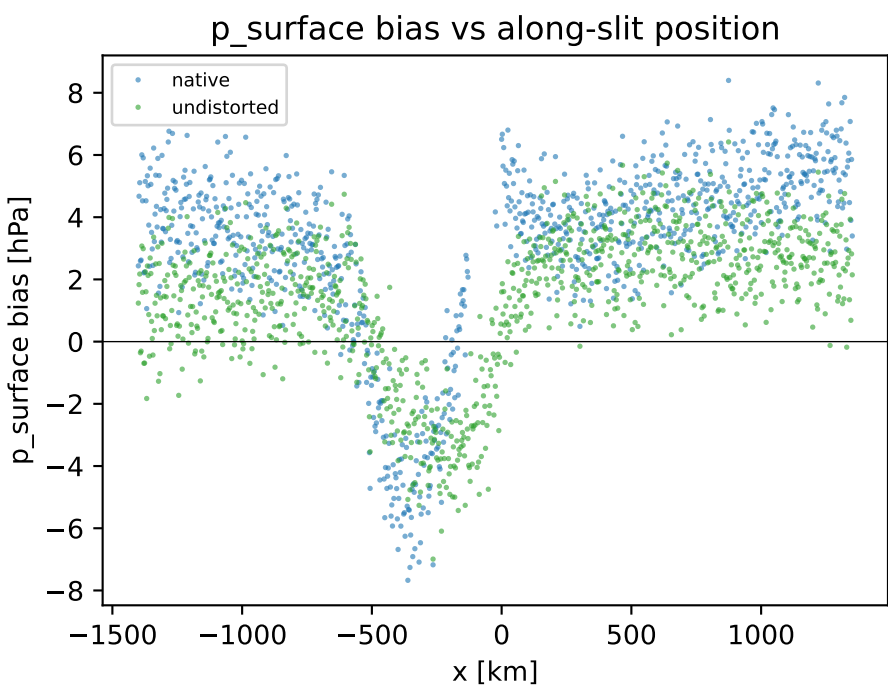
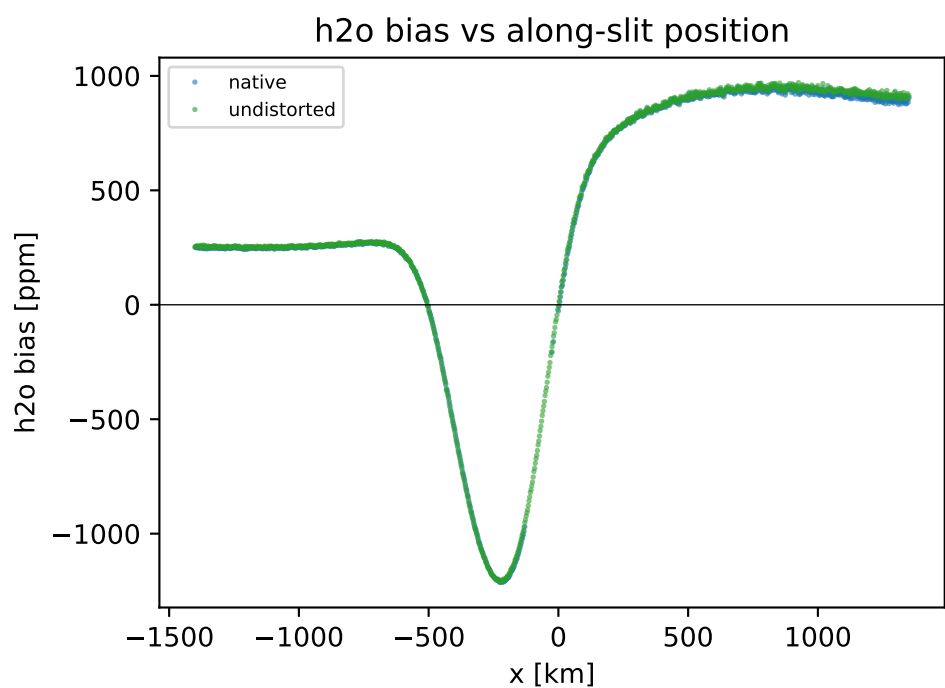
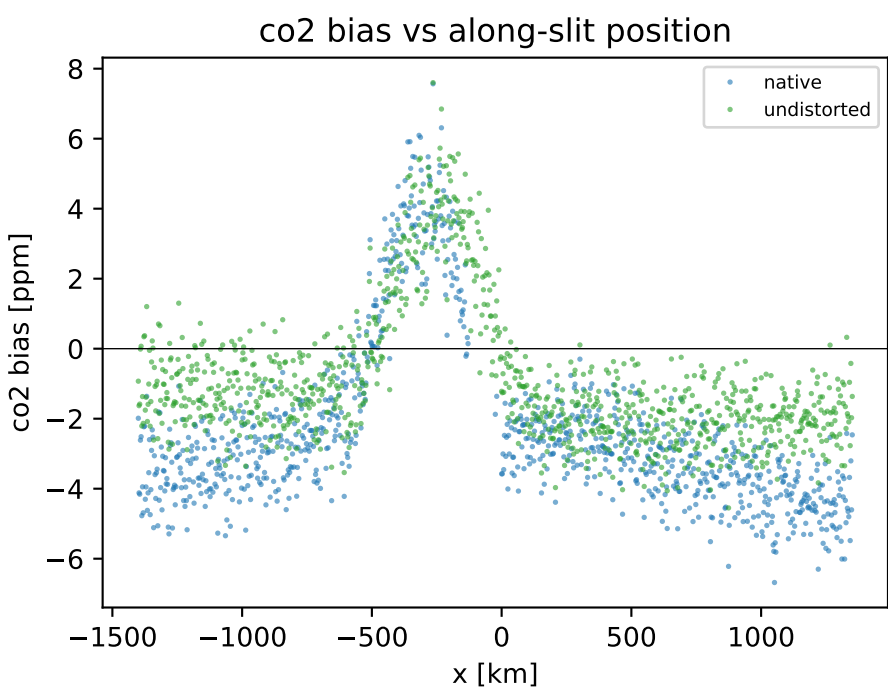
chi2_reduced vs along-slit position (outliers excluded)



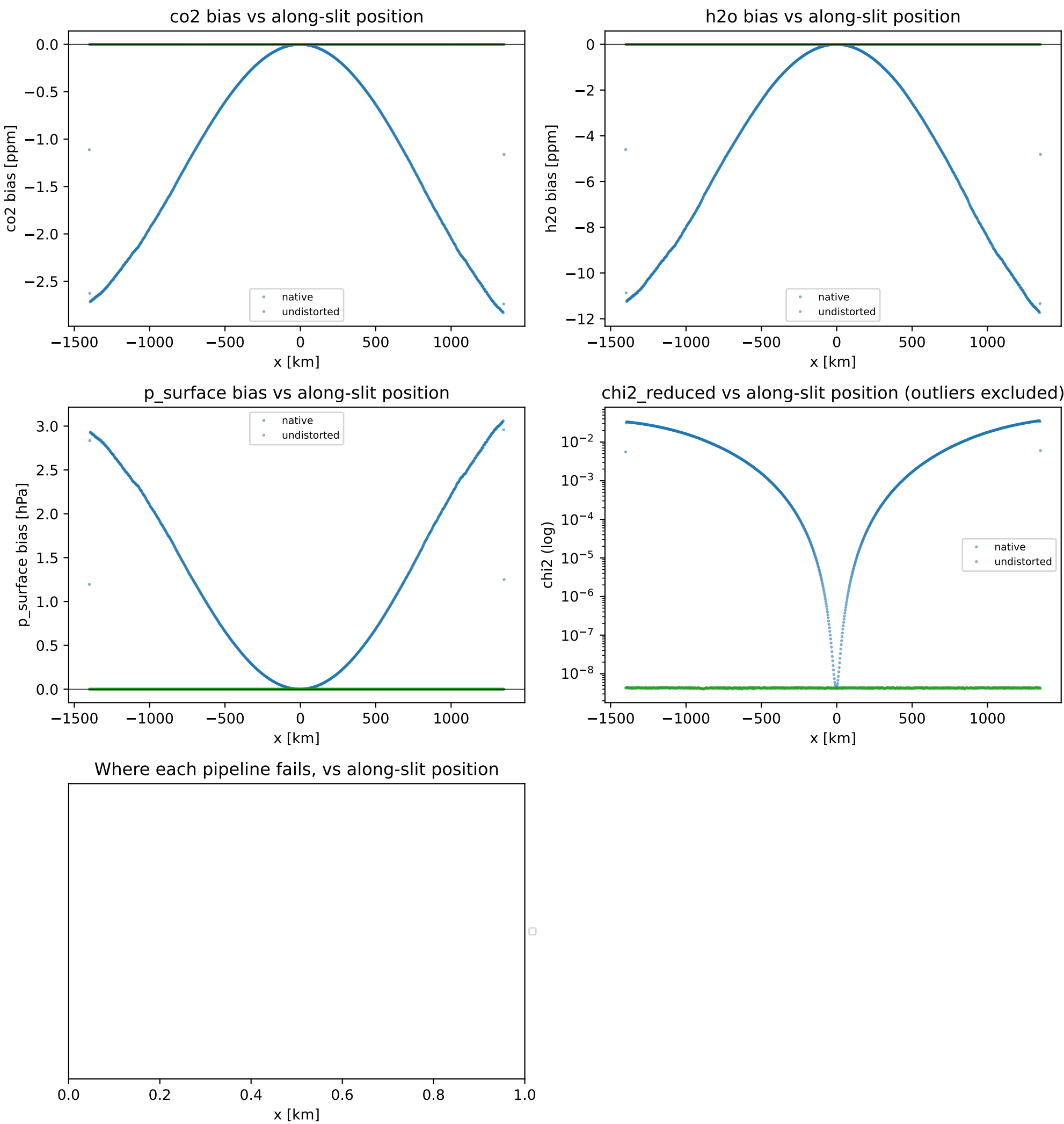
Where each pipeline fails, vs along-slit position



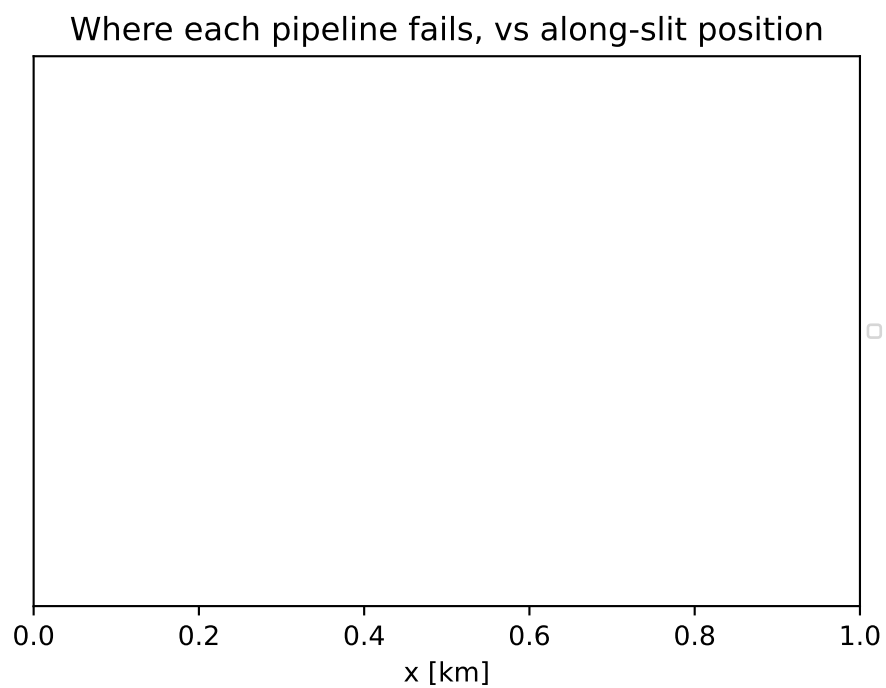
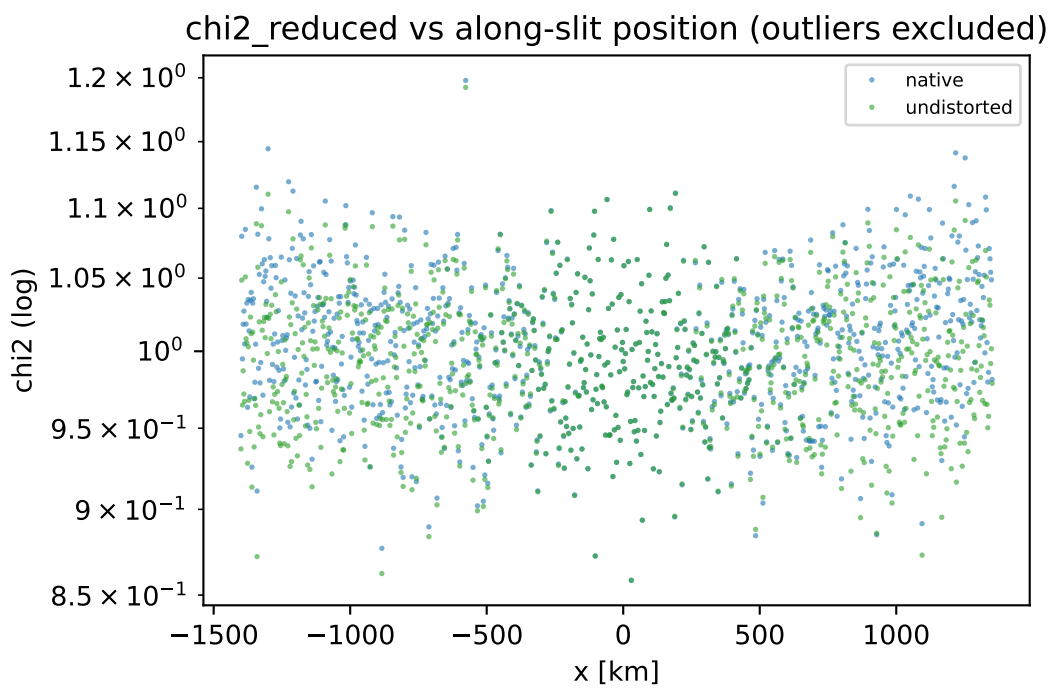
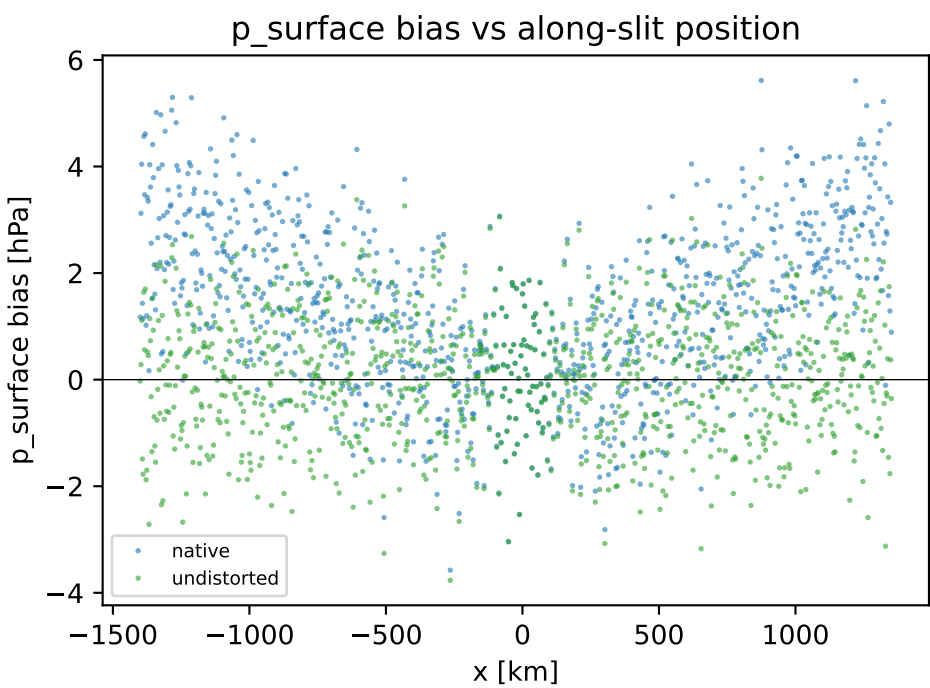
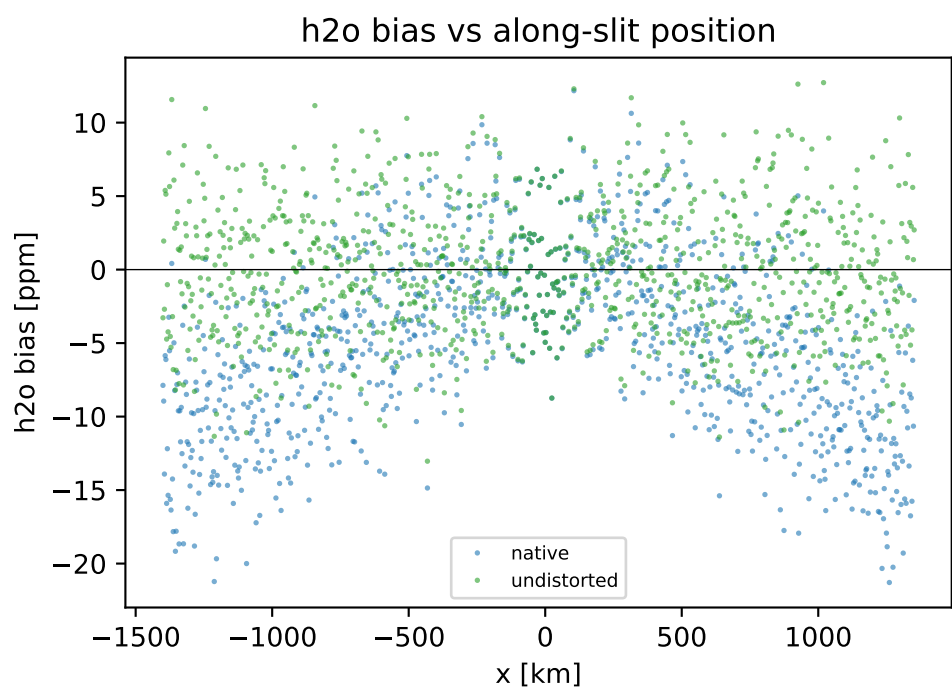
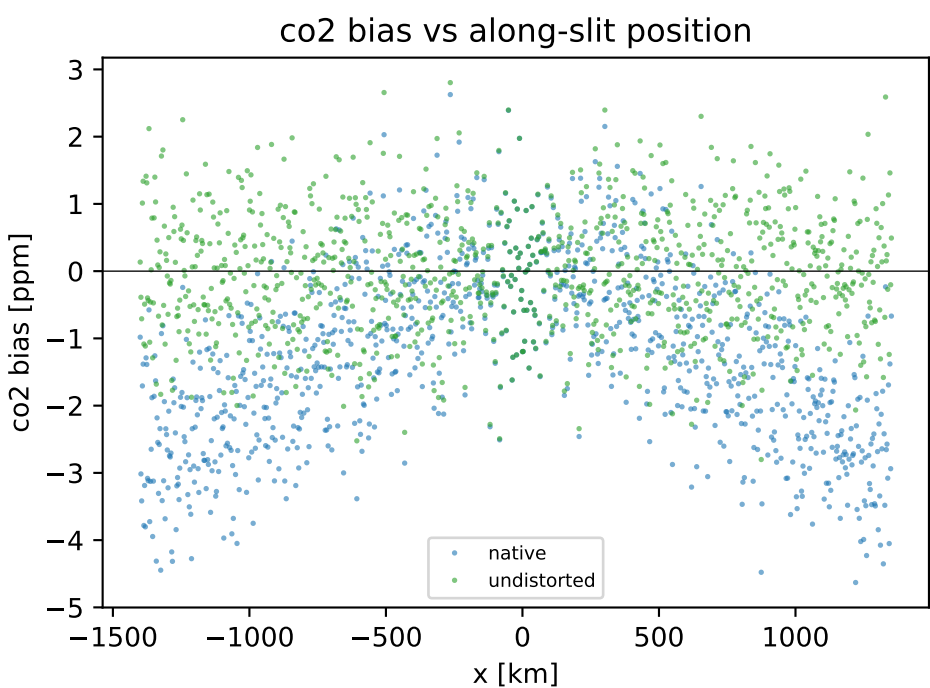
Joint FPA2 -- realistic, noise (native/undistorted)



Joint FPA2 -- uniform, no noise (native/undistorted)

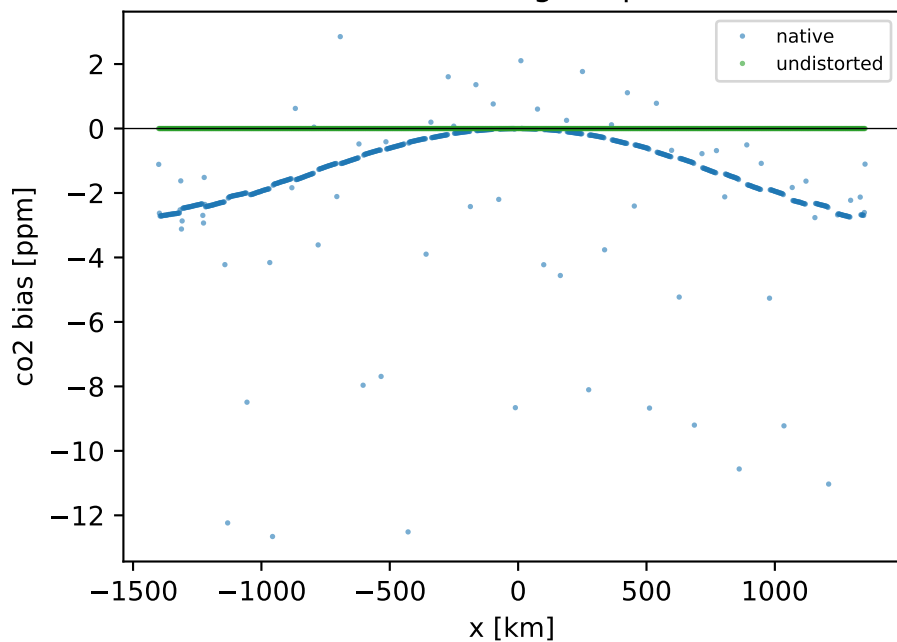


Joint FPA2 -- uniform, noise (native/undistorted)

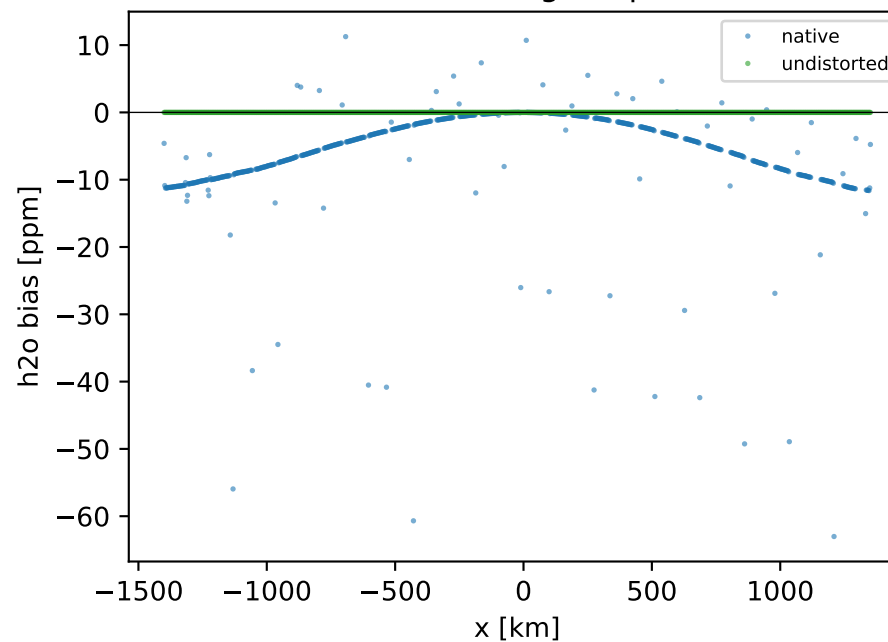


Joint FPA2 -- barcode, no noise (native/undistorted)

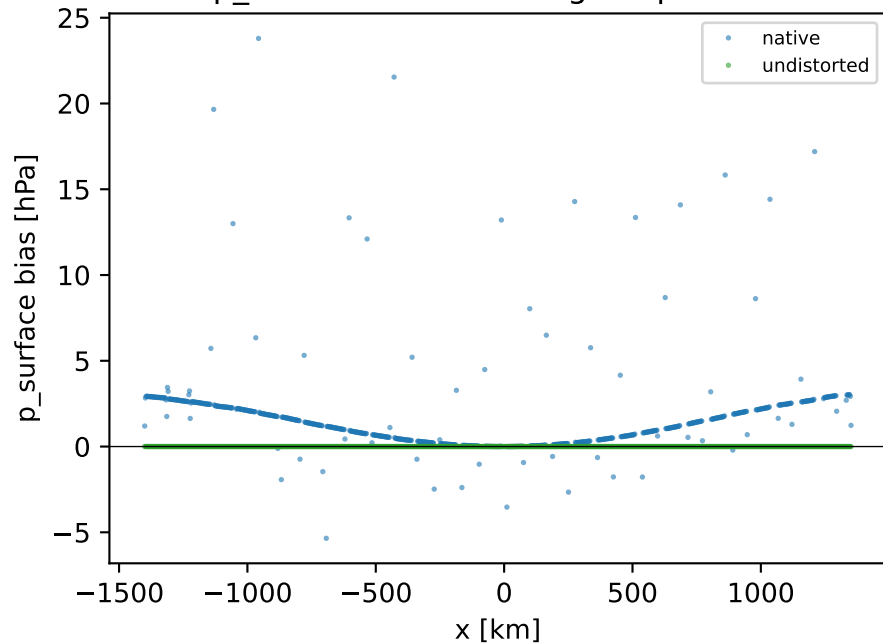
co2 bias vs along-slit position



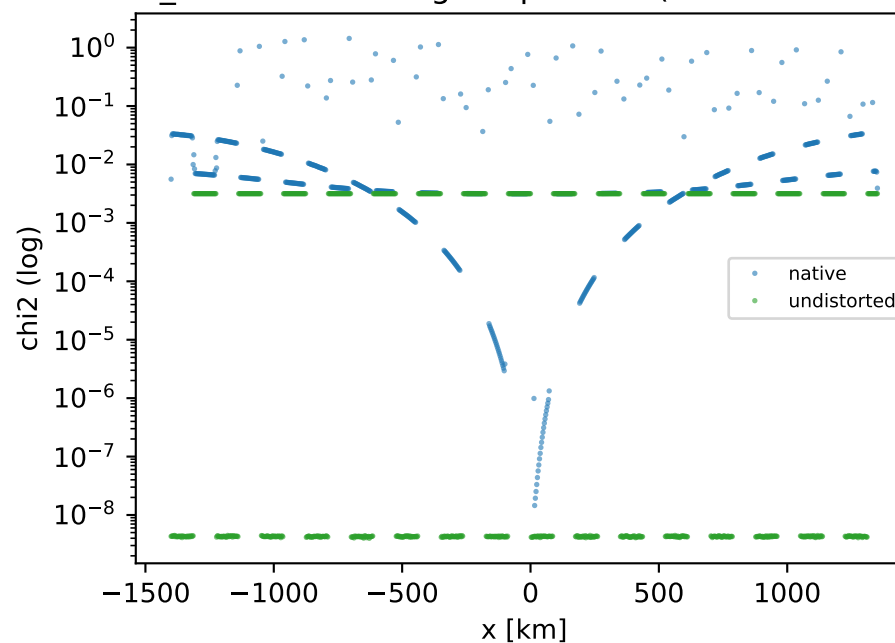
h2o bias vs along-slit position



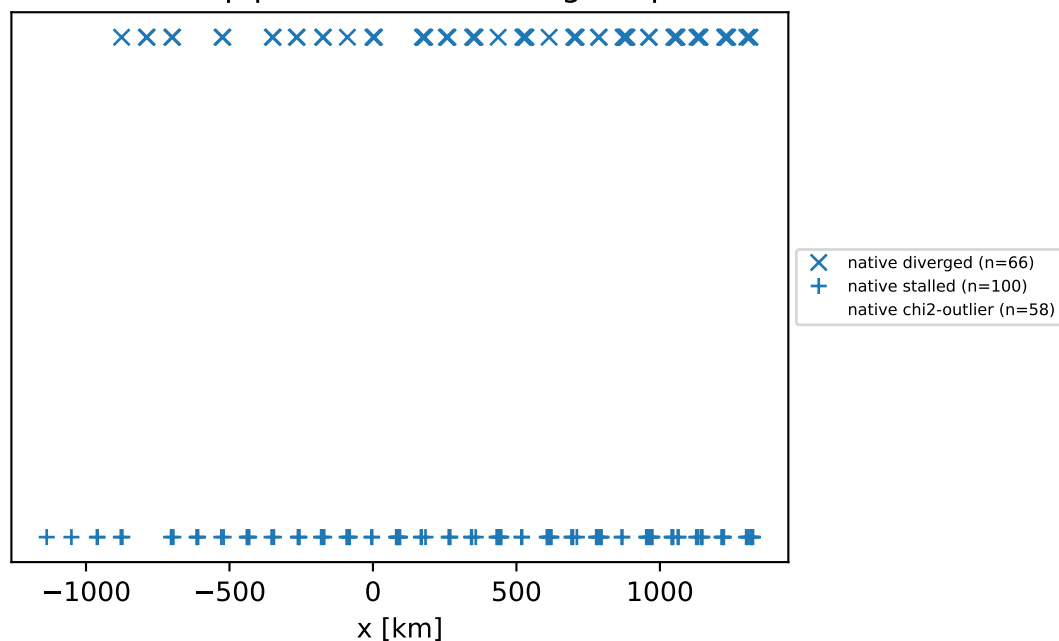
p_surface bias vs along-slit position



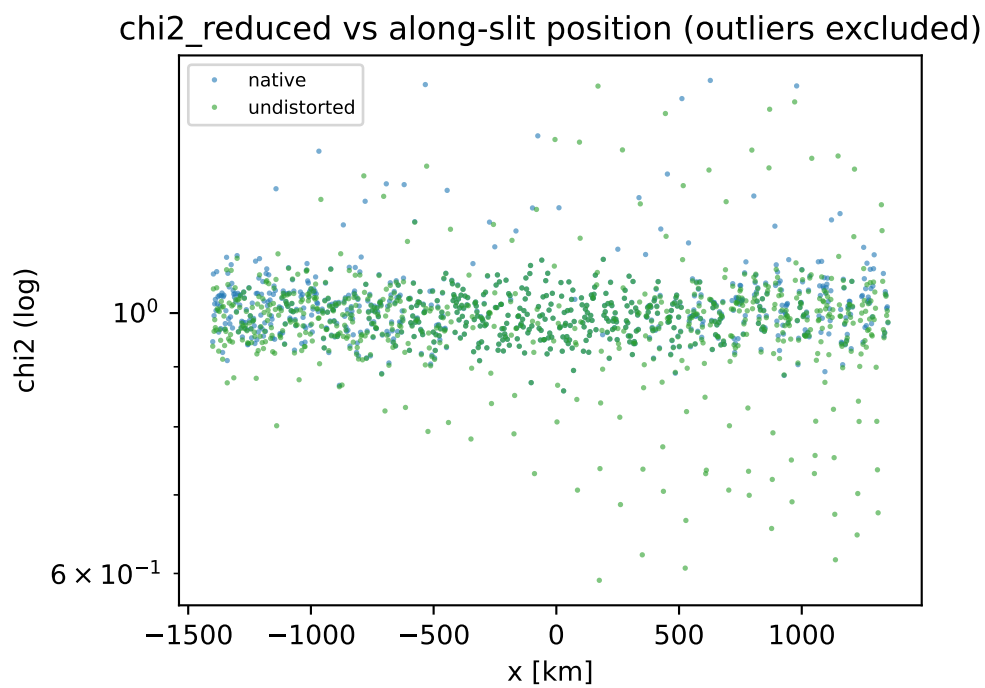
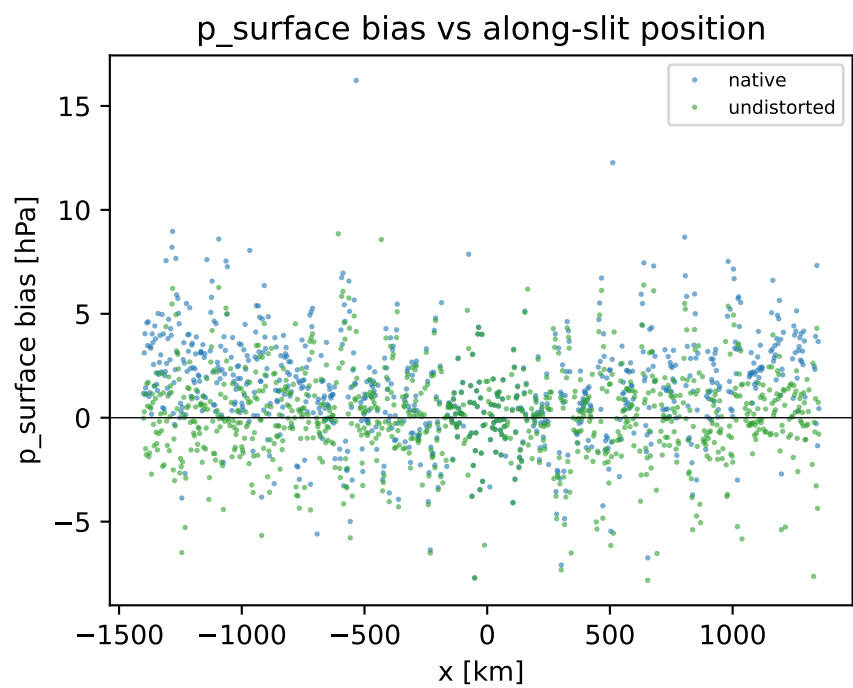
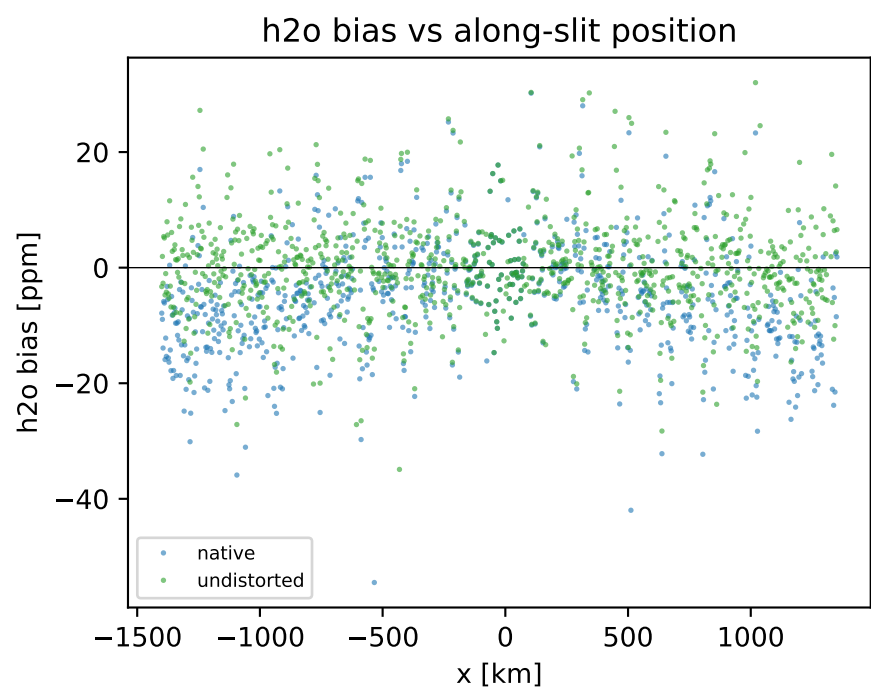
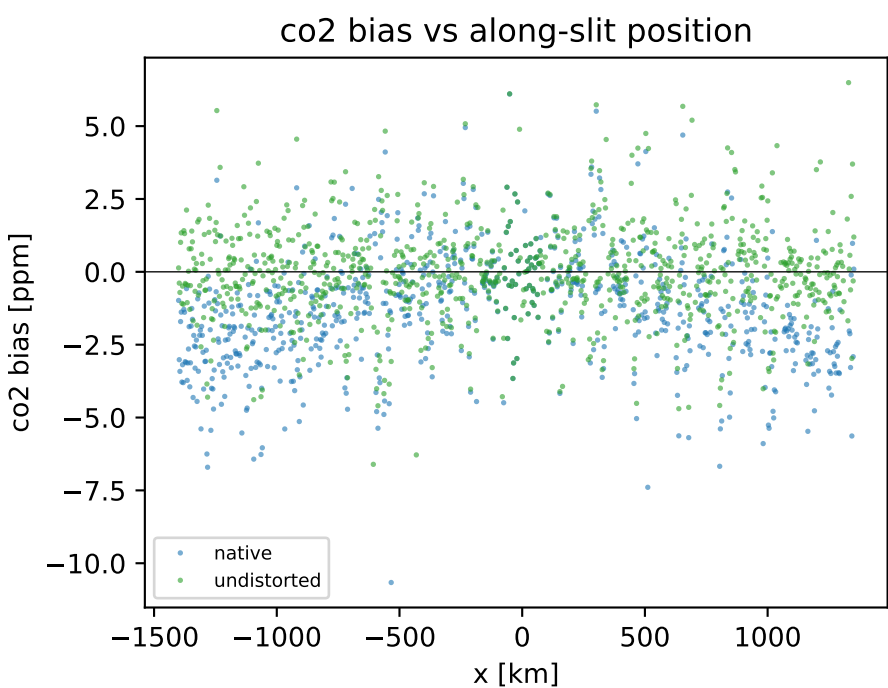
chi2_reduced vs along-slit position (outliers excluded)



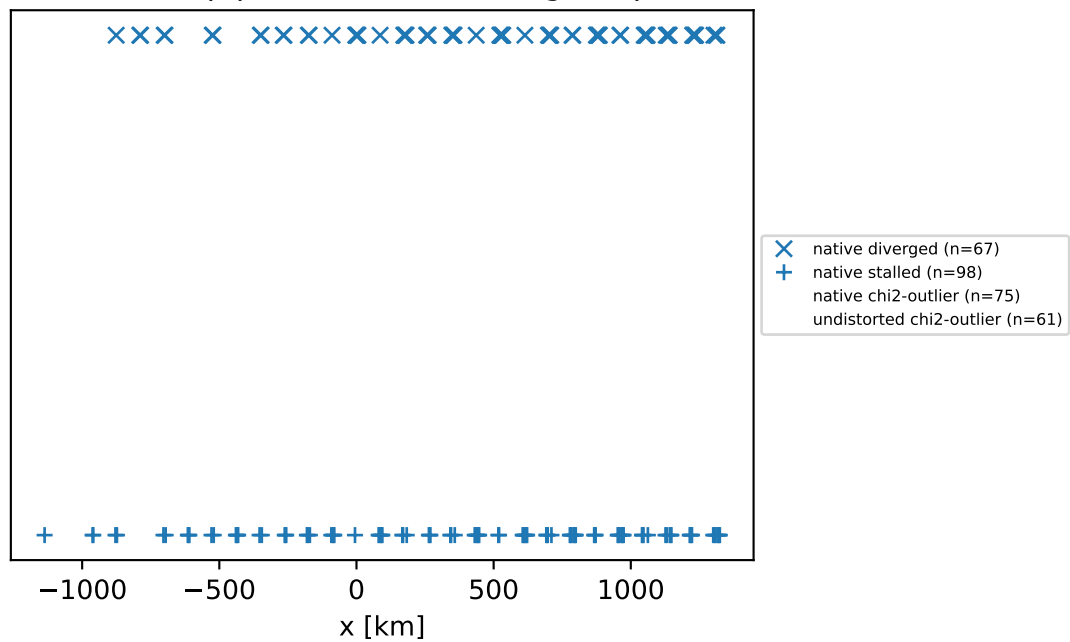
Where each pipeline fails, vs along-slit position



Joint FPA2 -- barcode, noise (native/undistorted)



Where each pipeline fails, vs along-slit position



FPA2 -- robust bias statistics across scenes/noise (native/undistorted)

scene	noise	quantity	pipeline	n	mean	std	median	MAD-sig	n_out
realistic	0	co2	native	1024	-2.219	2.251	-2.854	0.8827	14
realistic	0	h2o	native	1024	+354.7	615.2	+268.8	871	0
realistic	0	p_surface	native	1024	+3.028	2.723	+3.814	1.029	48
realistic	0	co2	undistorted	1024	-0.8539	1.79	-1.278	1.161	0
realistic	0	h2o	undistorted	1024	+362.9	618.3	+271.6	874.5	0
realistic	0	p_surface	undistorted	1024	+1.325	1.965	+1.579	1.847	0
realistic	1	co2	native	979	-2.289	2.442	-2.89	1.442	0
realistic	1	h2o	native	979	+395.5	595.2	+273.1	849.1	0
realistic	1	p_surface	native	979	+3.033	3.003	+3.773	1.941	0
realistic	1	co2	undistorted	1024	-0.8792	1.991	-1.422	1.28	0
realistic	1	h2o	undistorted	1024	+362.9	618.3	+271.9	875.8	0
realistic	1	p_surface	undistorted	1024	+1.37	2.284	+1.859	1.889	0
uniform	0	co2	native	1024	-1.209	0.921	-1.113	1.282	0
uniform	0	h2o	native	1024	-4.953	3.809	-4.507	5.288	0
uniform	0	p_surface	native	1024	+1.308	0.9942	+1.203	1.387	0
uniform	0	co2	undistorted	1024	-0.0006186	1.587e-05	-0.000617	1.582e-05	0
uniform	0	h2o	undistorted	1024	-0.002284	6.144e-05	-0.002287	5.948e-05	0
uniform	0	p_surface	undistorted	1024	+0.0005438	2.333e-05	+0.0005404	2.363e-05	0
uniform	1	co2	native	1024	-1.237	1.288	-1.196	1.393	0
uniform	1	h2o	native	1024	-5.034	5.962	-4.731	6.287	0
uniform	1	p_surface	native	1024	+1.355	1.55	+1.306	1.64	0
uniform	1	co2	undistorted	1024	-0.0285	0.8973	-0.04229	0.9185	0
uniform	1	h2o	undistorted	1024	-0.08389	4.396	-0.2735	4.457	0
uniform	1	p_surface	undistorted	1024	+0.04802	1.18	+0.06559	1.222	0
barcode	0	co2	native	800	-1.343	1.489	-1.115	1.311	3
barcode	0	h2o	native	800	-5.631	6.813	-4.596	5.492	5
barcode	0	p_surface	native	800	+1.566	2.259	+1.238	1.449	12
barcode	0	co2	undistorted	1024	-0.0005822	4.123e-05	-0.0005852	5.262e-05	0
barcode	0	h2o	undistorted	1024	-0.002175	0.0001347	-0.002194	0.0001605	0
barcode	0	p_surface	undistorted	1024	+0.0005812	4.641e-05	+0.0005775	5.816e-05	0
barcode	1	co2	native	784	-1.28	1.923	-1.268	1.728	0
barcode	1	h2o	native	784	-5.161	9.537	-4.76	8.311	0
barcode	1	p_surface	native	784	+1.474	2.526	+1.39	2.156	0
barcode	1	co2	undistorted	963	-0.04073	1.638	-0.0882	1.217	0
barcode	1	h2o	undistorted	963	+0.0484	8.249	-0.3845	6.474	0
barcode	1	p_surface	undistorted	963	+0.08495	2.177	+0.09496	1.679	0